6. Perform basic video processing operations on the captured video

Aim:

To perform basic video processing operations by reading a captured video in Python and displaying it in slow and fast motion using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2

import numpy as np



img=cv2.imread(r"C:\Users\Maruthi\Downloads\METAL.jpg",0)



if img is None:

    print("Image not found")

    exit()



f=np.fft.fft2(img)

fshift=np.fft.fftshift(f)
```

```
rows,cols=img.shape

crow,ccol=rows//2,cols//2


mask=np.ones((rows,cols),dtype=np.uint8)

mask[crow-5:crow+5,ccol-50:ccol-40]=0

mask[crow-5:crow+5,ccol+40:ccol+50]=0


filtered=fshift*mask


ishift=np.fft.ifftshift(filtered)

img_back=np.fft.ifft2(ishift)

img_back=np.abs(img_back)


img_back=cv2.normalize(img_back,None,0,255,cv2.NORM_MI
NMAX)

img_back=np.uint8(img_back)


cv2.imshow("Original Image",img)

cv2.imshow("Filtered Image",img_back)

cv2.waitKey(0)
```

cv2.destroyAllWindows()

Sample Input

Original Video

Sample Output

Video in slow motion and fast motion

Result:

The captured video was successfully read using Python and displayed in slow-motion and fast-motion using OpenCV.